

Warsaw University of Technology
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COMPUTER METHODS IN COMBUSTION

**SELF-IGNITION OF HYDROGEN
UNDER DIVERSE INITIAL
CONDITIONS**

**The effect of molar fuel composition on
ignition delay times**

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1 Introduction

The spontaneous ignition of a gas mixture without the presence of an external thermal energy source is known as autoignition. This process, however, does not occur instantaneously. The time interval that elapses between the initial stage of the chemical reactions and the rapid temperature spike necessary for sustained, autonomous flame propagation is defined as the ignition delay.

From the perspective of aerospace engineering and modern power technologies, analyzing the autoignition time of hydrogen plays a fundamental role in designing and optimizing stable combustion systems. The growing interest in hydrogen as a zero-emission fuel makes a thorough investigation of its thermodynamic stability limits under varying operating conditions a crucial research topic.

The objective of this project was to numerically investigate the autoignition process of hydrogen and determine the extent to which initial parameters dictate the ignition delay time. As part of the study, the influence of the mixture's structure was also evaluated by modifying the molar fuel quantity. Utilizing the Cantera framework, three distinct computational scenarios were investigated to evaluate the system's sensitivity to pressure fluctuations, initial temperature variations, and species proportions under both isothermal and isobaric reactor constraints.

A uniform detection criterion for the moment of ignition was applied across all simulations. The ignition delay was identified as the time required to achieve a temperature increase of more than 400 K relative to its initial value, which is mathematically described by the following relation:

$$t_{\text{ignition time}} = t_{\text{initial temperature}+400\text{K}}$$

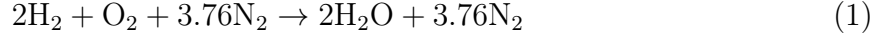
2 State of the Art

Numerical modeling of hydrogen auto-ignition is crucial for the development of clean energy and advanced propulsion technologies. The phenomenon is deeply non-linear and sensitive to initial thermodynamic properties, traditionally evaluated using zero-dimensional ideal gas reactor models under constant volume or constant pressure conditions. Numerical frameworks, most notably Cantera combined with detailed kinetic mechanisms like GRI-Mech 3.0, allow for accurate simulation of chemical kinetics and determination of ignition delay times. Extensive research indicates that the ignition delay of hydrogen exhibits a critical dependence on initial temperature and pressure, displaying non-monotonic behavior across different pressure regimes due to competition between radical chain-branching and termination reactions. Furthermore, the molar composition of the fuel-oxidizer mixture, specifically the equivalence ratio, significantly shifts the ignition thresholds. Modern computational studies focus on mapping these multi-variable dependencies to validate chemical solvers and optimize combustion stability in real-world applications.

3 Program description

The developed Python script utilizes the Cantera library and the GRI-Mech 3.0 (`gri30.yaml`) mechanism to simulate the transient autoignition of a hydrogen-air mixture. The global

stoichiometric combustion reaction in air is defined as follows:



3.1 Reactor Models and Architecture

The core functionality is handled by the `combust` function, which integrates zero-dimensional ideal gas reactor networks over a time domain with a discrete step of $\Delta t = 10^{-6}$ s. The script supports two distinct configurations:

- **Constant Volume Mode (`IdealGasReactor`):** Models a closed system where volume remains constant, causing pressure and temperature to rise due to exothermic reactions.
- **Constant Pressure Mode (`IdealGasConstPressureReactor`):** Models an isobaric environment where the system expands to maintain ambient pressure.

3.2 Ignition Criterion and Parametric Studies

The autoignition delay time (`ignition_delay`) is defined as the time required for the reactor temperature to increase by more than 400 K above its initial value:

$$T_{\text{reactor}} \geq T_{\text{initial}} + 400 \text{ K} \quad (2)$$

The program performs three parameter sweeps to evaluate ignition delay dependency:

1. Pressure variation: Investigates a stoichiometric mixture at 1250 K with pressure ranging from 1 to 5 atm using the constant volume reactor.
2. Temperature variation: Investigates the mixture at 5 atm with initial temperatures from 1012.5 K to 1450 K using the constant pressure reactor.
3. Composition variation: Analyzes ignition delay as a joint function of fuel quantity (0.1 to 0.8 moles of H_2) and pressure (1 to 5 atm).

4 Results

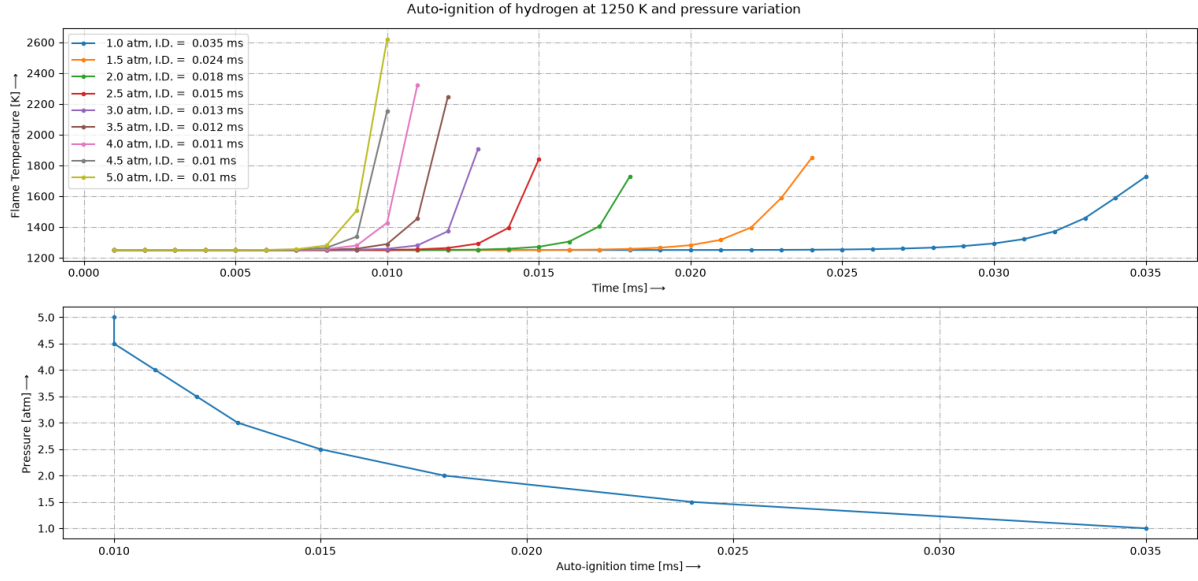


Figure 1: Effect of initial pressure variation on hydrogen autoignition at 1250 K (top: temperature profiles, bottom: ignition delay time).

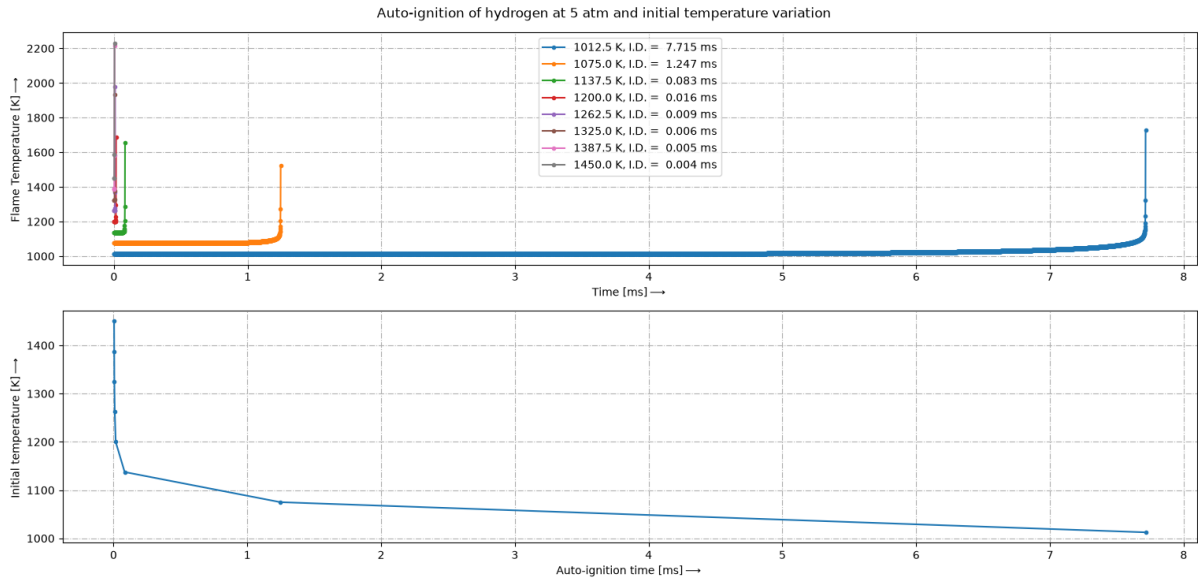


Figure 2: Effect of initial temperature variation on hydrogen autoignition at 5 atm (top: temperature profiles, bottom: ignition delay time).

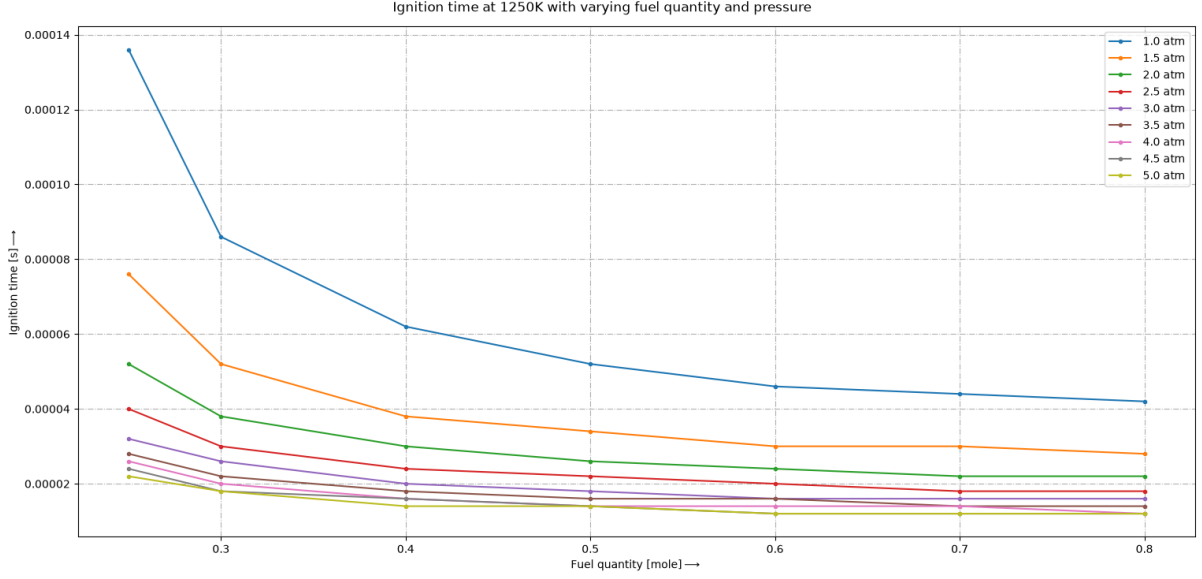


Figure 3: Ignition delay time as a function of H_2 fuel quantity across different pressure levels at 1250 K.

5 Conclusions

Based on the transient numerical simulations conducted for hydrogen autoignition using the Cantera framework, the following conclusions can be formulated:

1. **Impact of Initial Pressure:** An increase in initial pressure under a constant initial temperature (1250 K) significantly reduces the ignition delay time. Higher pressure increases the volumetric concentration of the reactants and the frequency of molecular collisions, which accelerates the radical chain-branching reactions responsible for the onset of autoignition.
2. **Impact of Initial Temperature:** The ignition delay time exhibits an exponential-like sensitivity to the initial temperature. At a constant pressure of 5 atm, lower temperatures prolong the induction period.
3. **Impact of Mixture Composition:** The parametric study involving the variation of fuel quantity (H_2 moles from 0.1 to 0.8) shows that the optimal (shortest) ignition delay is highly dependent on the fuel fraction. Extremely lean or rich mixtures tend to delay the ignition process, whereas the minimum induction time aligns closely with the regions where the availability of fuel and oxidizer optimizes the chain-initiation rates. This composition effect is consistently shifted toward shorter times as the overall system pressure increases.

References

- [1] MKWS26info - Accessed: 13.06.2026
- [2] Cantera documentation. <https://cantera.org/documentation/index.html> - Accessed: 13.06.2026